Non-Electrical Components in KiCad

PCBs are rarely, if ever, standalone elements. They are typically integrated with mechanical constraints and non-electrical components such as screws, chassis, and optical equipment.

It is good practice to include these components in your design: in the Bill of Materials (BOM), in the schematic for reference, and as footprints in the PCB layout to help prevent collisions with other components or traces. Including them in the 3D preview also gives you a better sense of how the final product will look.

In this KiCad tutorial, we will learn how to manage these components in an organized and efficient way.

Solution

To properly account for non-electrical components in your PCB design, create dedicated symbols and associate them with custom footprints that include embedded 3D models.

These elements should be placed on non-copper layers (e.g., user drawing layers) to ensure they do not interfere with the electrical design.

This approach ensures that mechanical components are considered at every stage, including the BOM, PCB layout, and 3D visualization.

What You'll Learn

In this tutorial, you will learn how to:

- Design a non-electrical symbol representing a mechanical component.
- Link that symbol to a custom footprint with an associated 3D model.
- Place the footprint on a non-electrical layer (such as a user drawing layer) to keep it separate from copper elements.

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1. Create a Symbol to Represent the Component

First, we will create a symbol representing our mechanical component. This allows us to place it in the schematic and include it in the BOM.

For this example, we will use the CP35/M, an optical cage plate.

Steps

1. Open **Symbol Editor** (Tools → Symbol Editor).
2. Create a new library or add an existing one.

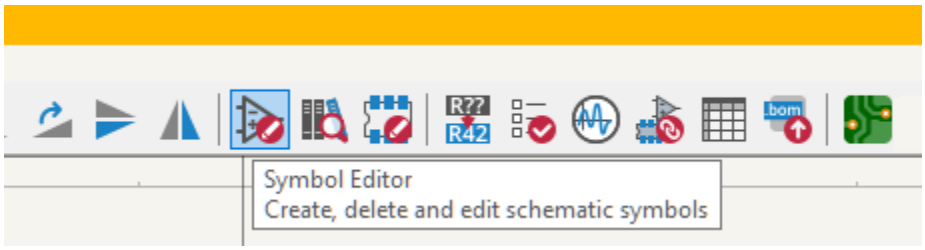


Figure 1: Open Symbol Editor

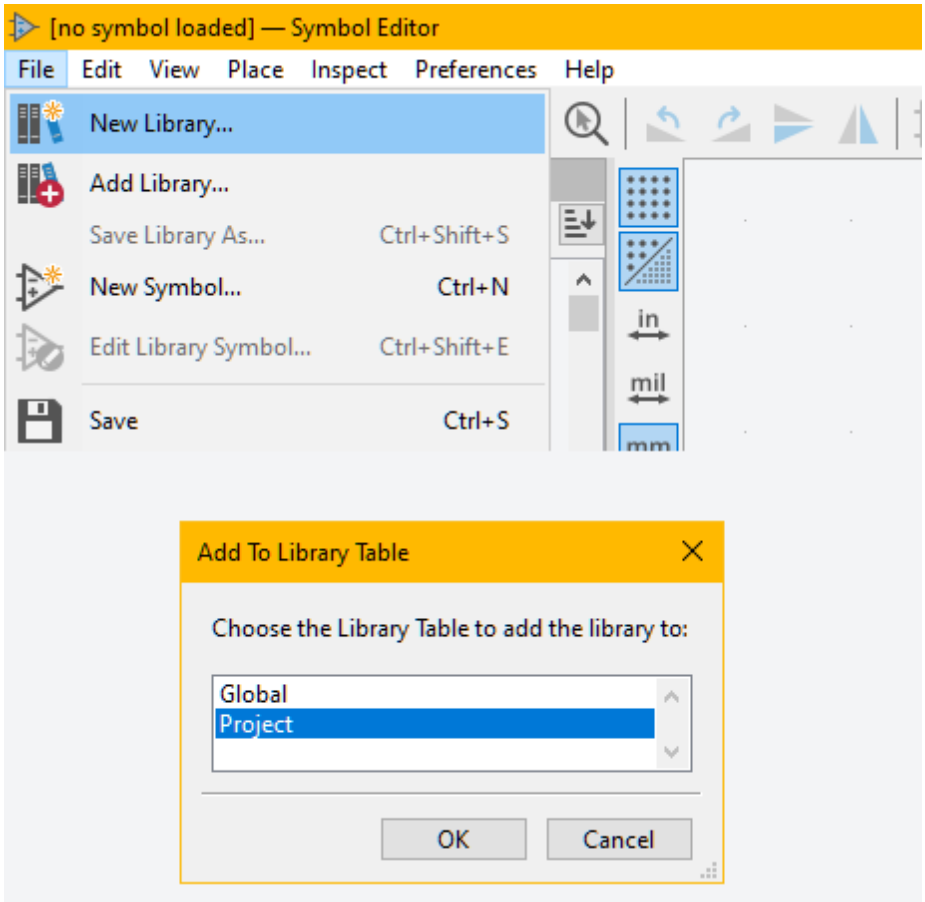


Figure 2: New Library

3. Choose whether the library should be **Global** or **Project** specific.
4. Name the symbol library, for example, `Optical_Components.kicad_sym`.
5. Click **New Symbol**.

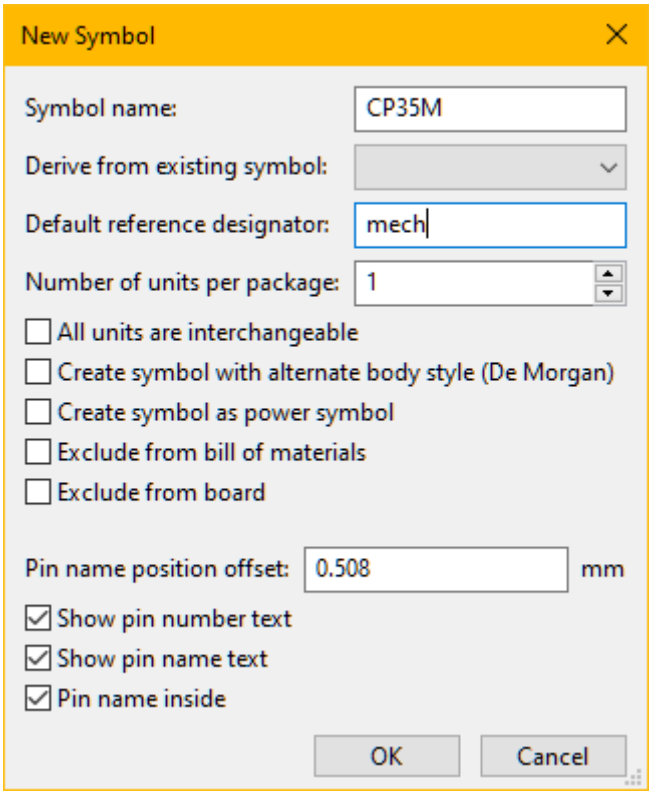


Figure 3: New Symbol

6. Name the symbol `CP35M`.
7. Assign a reference designator, for example, `MECH`.
8. Open **Symbol Properties** from the menu bar.
9. Configure the symbol properties:
   - Set the symbol value (it is not particularly important at this stage).
   - Enable **Exclude from simulation** (optional).
   - Add a link to the component datasheet, if available.
   - Provide a brief description.
   - Create a custom field, for example, `Link`, and paste the component's website URL. This information can be useful in the BOM.
10. Save the symbol, keeping the Symbol Editor open.

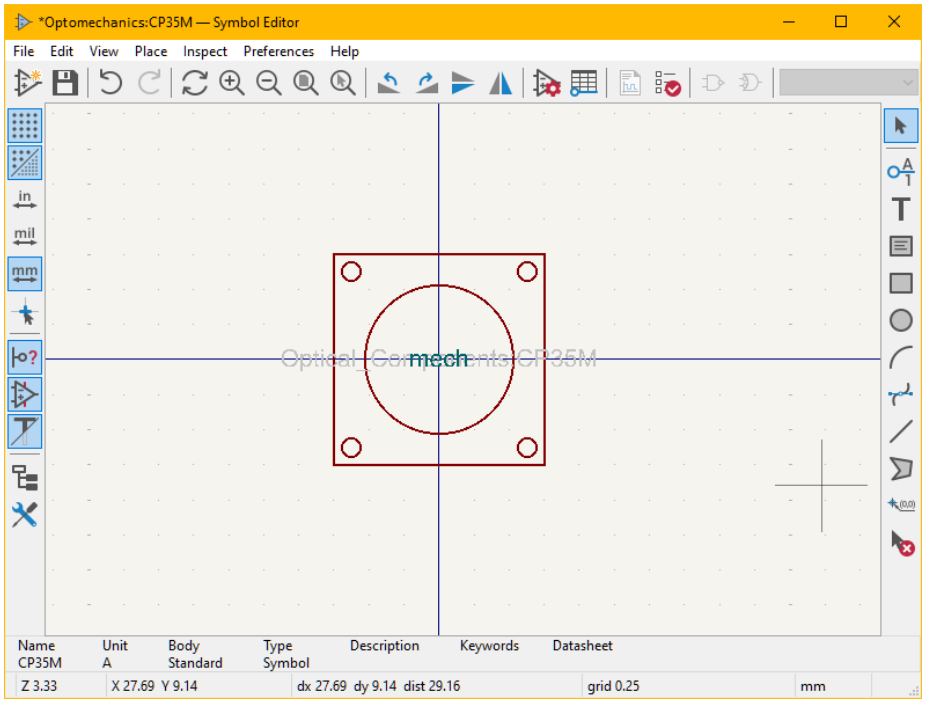


Figure 4: Symbol Editor

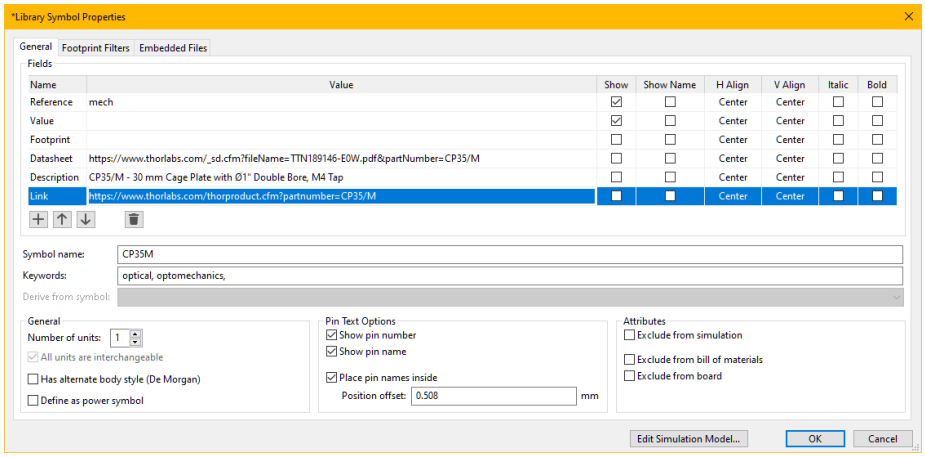


Figure 5: Symbol Properties

Note: This symbol acts purely as a placeholder and does not affect the electrical functionality of the circuit.

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2. Create a Footprint

Next, we will create a custom footprint for the mechanical component.

Steps

1. Open **Footprint Editor**.

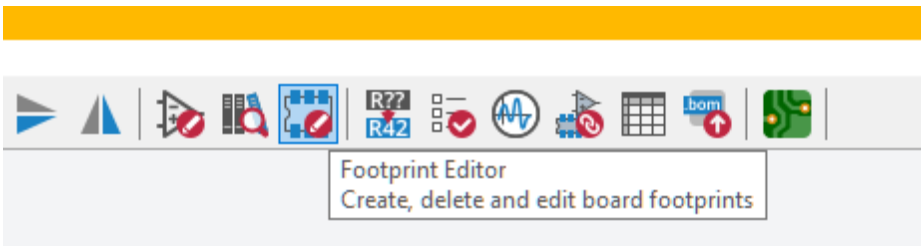


Figure 6: Open Footprint Editor

2. Create a new library or add an existing one.
3. Choose whether the library should be **Global** or **Project** specific.
4. Name the footprint library, for example, `Optomechanics.pretty`.
5. Click **New Footprint** (File → New Footprint).
6. Name the footprint, for example, `CP35M`, and save it in the library you just created (`Optomechanics.pretty`).
7. Go to **File → Import → Graphics** and select your DXF file. For this example, we use the `User.Drawings` layer.
8. You may want to clean up the imported DXF geometry and center it.
9. Open **Footprint Properties** from the menu bar.
10. Configure the footprint properties:
    - Optionally, add a datasheet link.
    - Provide a brief description.
    - Add the path to the STEP file and the component website, if applicable.
    - Set **Component Type** to `Unspecified`.
    - Enable **Exclude from position files**.
    - Enable **Do not populate**.

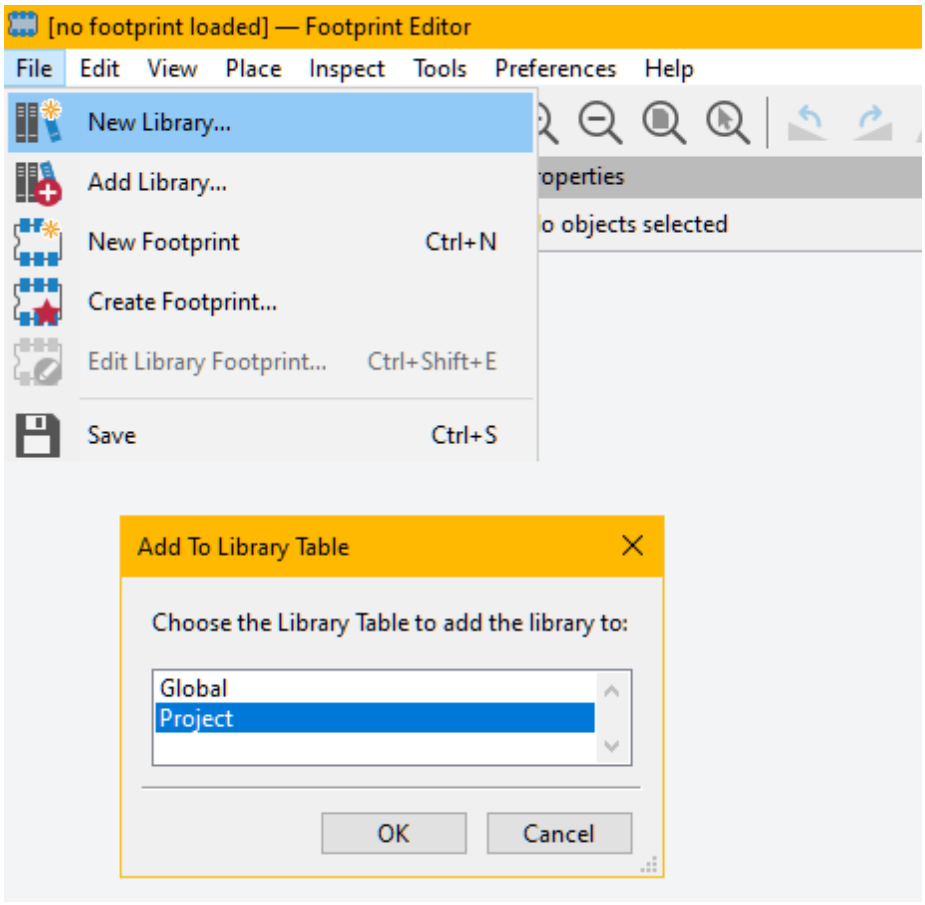


Figure 7: New Footprint Library

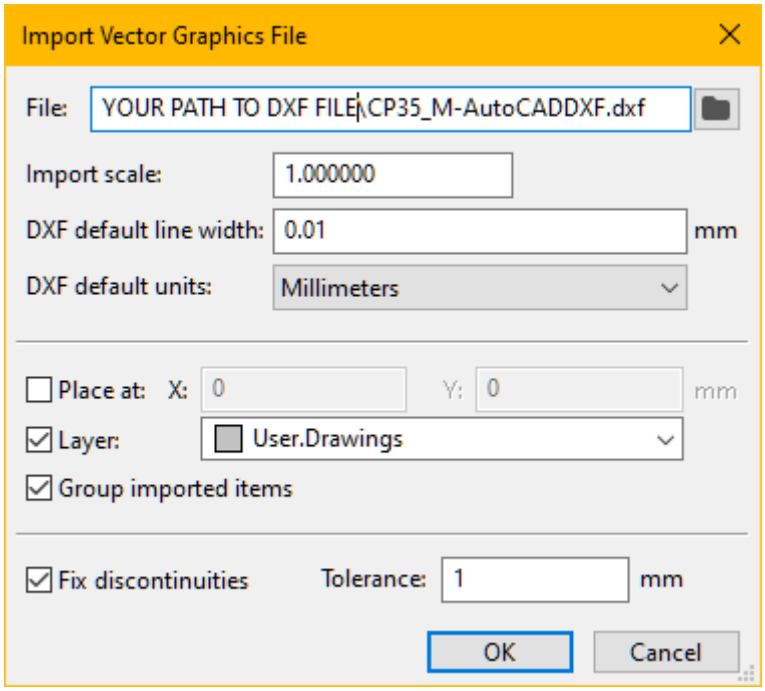


Figure 8: Import Graphics

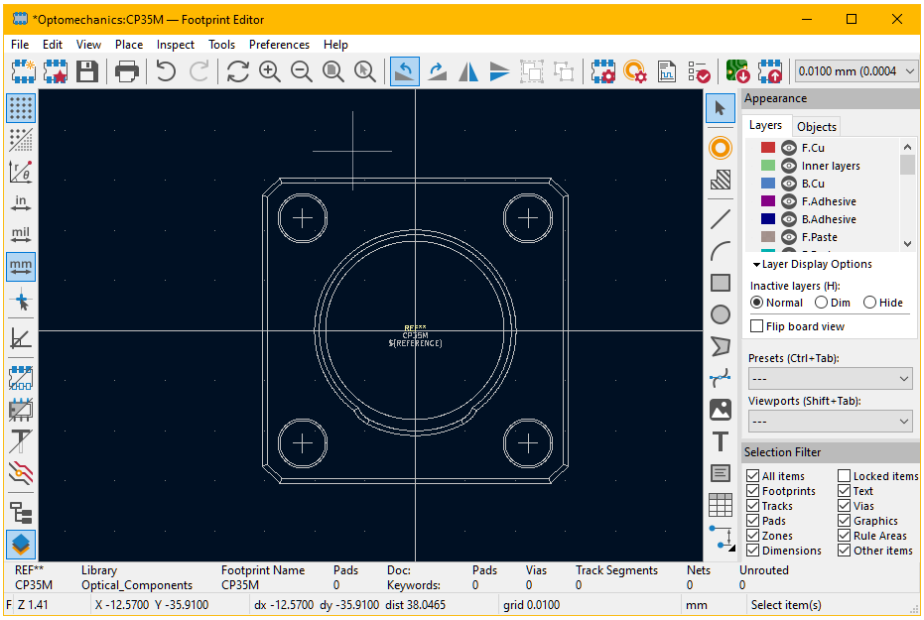


Figure 9: Imported DXF

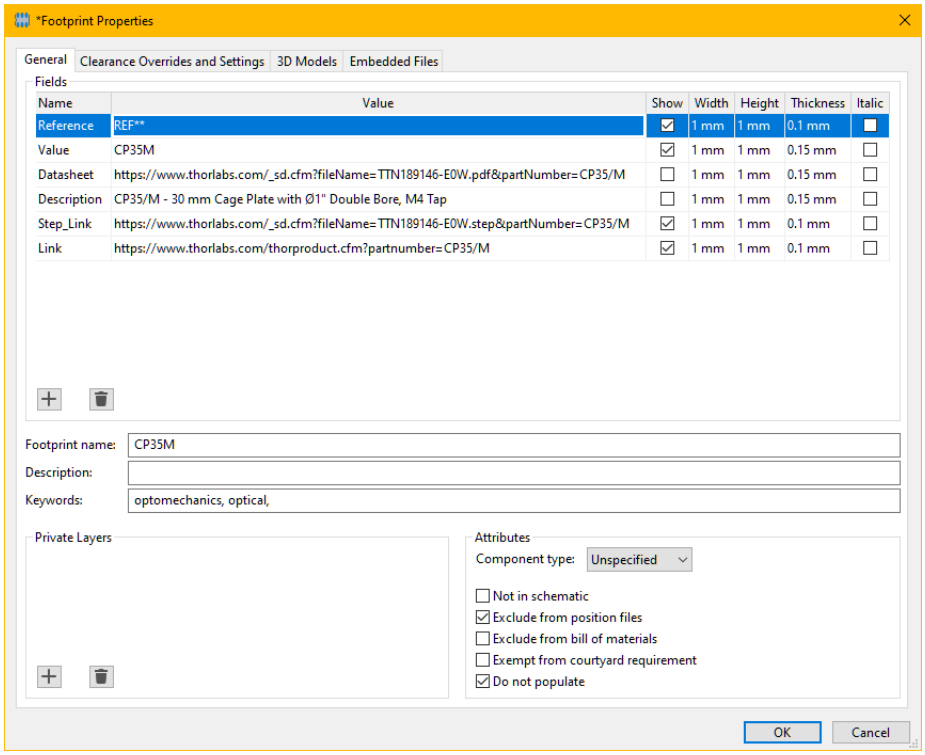


Figure 10: Footprint Properties - General

11. Open the **3D Models** tab.

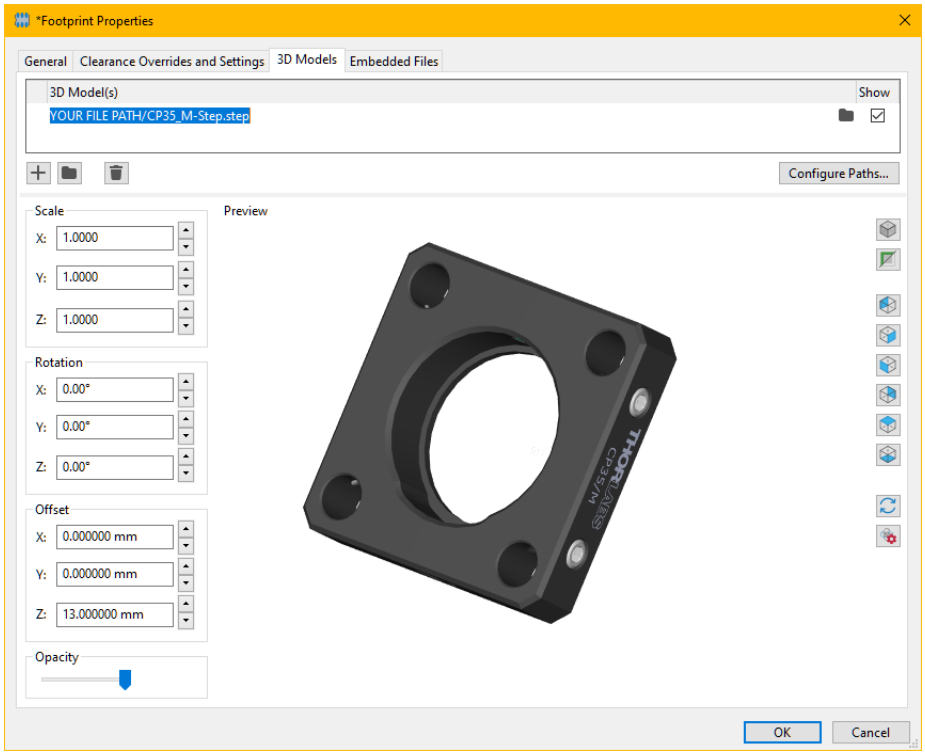


Figure 11: Footprint Properties - 3D Models

12. Click the + button to add a new model path and select your `.step` file.
13. Make sure the **Show** option on the right is checked.
14. Click **OK** and save the footprint.

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3. Link the Symbol to the Footprint

Now we need to associate the symbol created in the first step with the footprint.

1. Return to the **Symbol Editor**.
2. Open **Symbol Properties**.
3. Locate the footprint field and search for the footprint you just created.
4. Save the symbol and place it in your schematic.

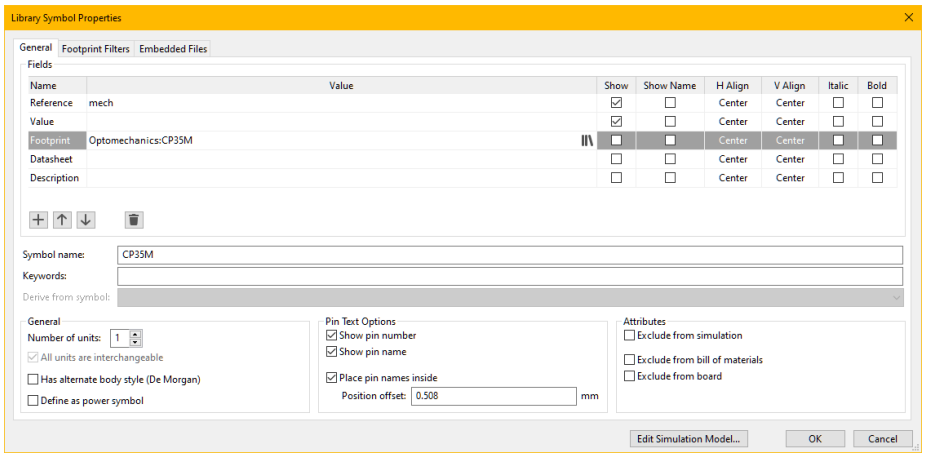


Figure 12: Footprint Assignment

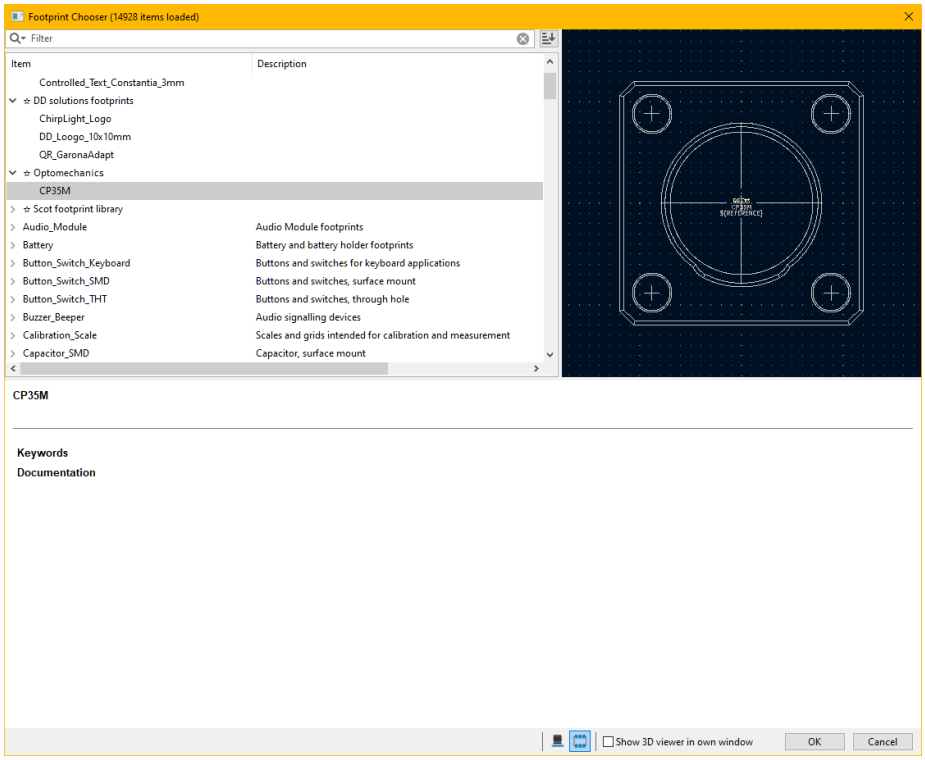


Figure 13: Footprint Assignment - Footprint Chooser

4. Final Result

You can now use the mechanical component throughout your design.

- The BOM will contain the component and its associated information.
- The PCB layout will display the DXF geometry on the `User.Drawings` layer.
- The 3D viewer will include the associated mechanical model.

This makes it easier to account for mechanical constraints and interactions during PCB design, helping to prevent potential collisions and integration issues.

Download

You can download the KiCad library containing the optical components here:

Optical Components – KiCad Library